WORKSHEET 1.5

Resolving 0/0: Factoring, Conjugates and Compound Fractions

Unit 1: Limits and Continuity

Key ideas

- $\frac{0}{0}$ is **indeterminate**: a signal to rewrite, never an answer.
- Polynomials: factor and cancel. Square roots: multiply by the conjugate. Fractions within fractions: use a common denominator.
- Absolute values and piecewise rules: take one-sided limits separately.
- Keep "lim" on every line that still contains the variable; drop it on the line where you substitute.

Worked examples

Example 1 · Factoring

Evaluate $\lim_{x\to 2} \dfrac{x^2+3x-10}{x^2-4}$.

Solution. $\lim_{x\to 2} \dfrac{(x+5)(x-2)}{(x-2)(x+2)} = \lim_{x\to 2} \dfrac{x+5}{x+2}$.

The limit is $\dfrac{7}{4}$.

Example 2 · Conjugate

Evaluate $\lim_{x\to 0} \dfrac{\sqrt{x+25}-5}{x}$.

Solution. Multiply by $\dfrac{\sqrt{x+25}+5}{\sqrt{x+25}+5}$: $\dfrac{x}{x\left(\sqrt{x+25}+5\right)} = \dfrac{1}{\sqrt{x+25}+5}$.

The limit is $\dfrac{1}{10}$.

Example 3 · Compound fraction

Evaluate $\lim_{x\to 3} \dfrac{\frac{1}{x}-\frac{1}{3}}{x-3}$.

Solution. $\dfrac{1}{x}-\dfrac{1}{3} = \dfrac{3-x}{3x}$, so the expression is $\dfrac{3-x}{3x(x-3)} = -\dfrac{1}{3x}$.

The limit is $-\dfrac{1}{9}$.

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Practice

CORE

1. Evaluate $\lim_{x\to -1} \dfrac{x^2-1}{x+1}$.

2. Evaluate $\lim_{x\to 3} \dfrac{x^2-2x-3}{x-3}$.

3. Evaluate $\lim_{h\to 0} \dfrac{(4+h)^2-16}{h}$.

4. Evaluate $\lim_{x\to 1} \dfrac{x^3-1}{x-1}$.

EXAM LEVEL

5. Evaluate $\lim_{x\to 25} \dfrac{\sqrt{x}-5}{x-25}$.

6. Evaluate $\lim_{x\to 1} \dfrac{\frac{1}{x+2}-\frac{1}{3}}{x-1}$.

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7. Find the value of k for which $\lim_{x\to -2} \dfrac{x^2+kx+6}{x+2}$ exists, and find the limit.

8. Evaluate $\lim_{x\to 5^-} \dfrac{|x-5|}{x^2-25}$.

Question 9 · Free-response practice

Let $f(x) = \dfrac{x^2-9}{\sqrt{x+1}-2}$ for $x \neq 3$, $x \geq -1$.

(a) Show that direct substitution at $x = 3$ gives an indeterminate form.

(b) Evaluate $\lim_{x\to 3} f(x)$, showing correct limit notation at each step.

(c) What value must be assigned to $f(3)$ to make f continuous at $x = 3$? Give a reason.

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Worked solutions

1. $x - 1 \to -2$.
2. $\dfrac{(x-3)(x+1)}{x-3} = x + 1 \to 4$.
3. $\dfrac{8h + h^2}{h} = 8 + h \to 8$.
4. $x^2 + x + 1 \to 3$.
5. $\dfrac{\sqrt{x} - 5}{(\sqrt{x} - 5)(\sqrt{x} + 5)} = \dfrac{1}{\sqrt{x} + 5} \to \dfrac{1}{10}$.
6. $\dfrac{3 - (x+2)}{3(x+2)(x-1)} = \dfrac{-(x-1)}{3(x+2)(x-1)} = -\dfrac{1}{3(x+2)} \to -\dfrac{1}{9}$.
7. The numerator must vanish at -2: $4 - 2k + 6 = 0$, so $k = 5$. Then $x^2 + 5x + 6 = (x+2)(x+3)$ and the limit is 1.
8. For $x < 5$, $|x - 5| = -(x - 5)$, so the expression is $-\dfrac{1}{x+5} \to -\dfrac{1}{10}$.
9. (a) Numerator: $9 - 9 = 0$. Denominator: $\sqrt{4} - 2 = 0$. The form is $\frac{0}{0}$.

   (b) $\lim\limits_{x \to 3} \dfrac{(x^2 - 9)\left(\sqrt{x+1} + 2\right)}{(x+1) - 4} = \lim\limits_{x \to 3} \dfrac{(x-3)(x+3)\left(\sqrt{x+1} + 2\right)}{x - 3} = \lim\limits_{x \to 3} (x+3)\left(\sqrt{x+1} + 2\right) = 6 \cdot 4 = 24$.

   (c) $f(3) = 24$, because continuity at 3 requires $f(3) = \lim\limits_{x \to 3} f(x)$.

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